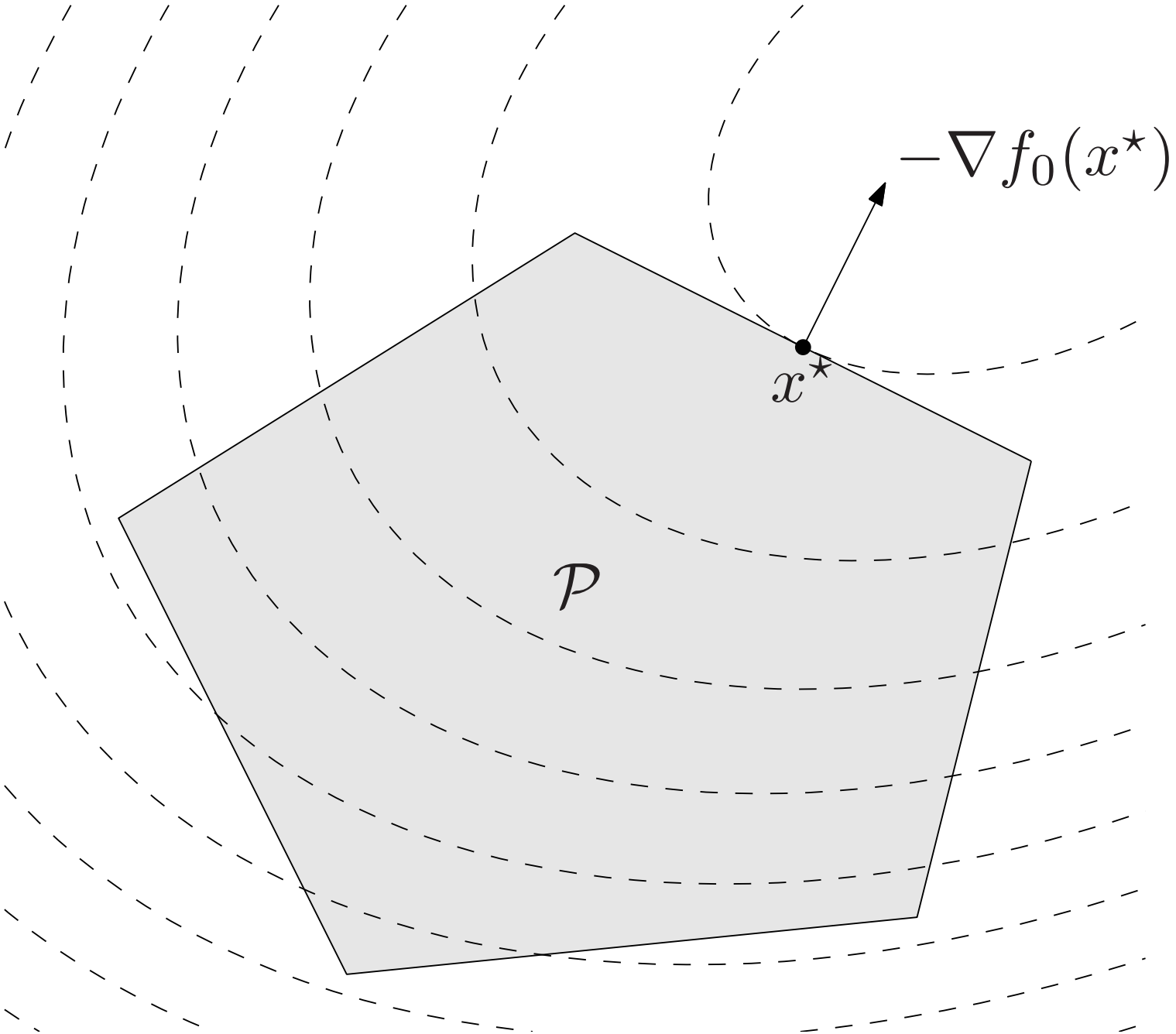

$-\nabla f_0(x^\star)$
$x^\star$
$\mathcal{P}$